

Hierarchical Models and Pooling Strategies

Hierarchical models Also known as *multilevel*, *mixed effects*, or *random effects* models, are designed for data with group or nested structure. These terms are often used interchangeably in the literature, though some authors distinguish them to emphasize subtle modeling differences.

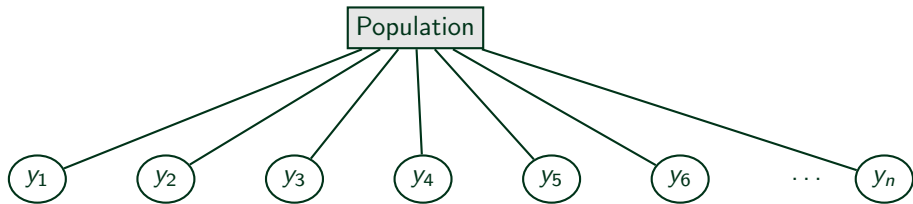
Pooling Strategies in Inference:

- **Complete pooling:** Assumes all groups share the same parameters. Ignores group-specific variation.
- **No pooling:** Each group is modeled separately with no shared information. Estimates may be unstable for small groups.
- **Partial pooling (Hierarchical):** Group-specific parameters are drawn from a common distribution, allowing the model to share information across groups.

Why Use Hierarchical Models?

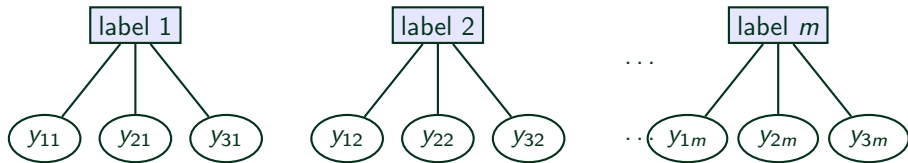
They balance individual group estimates with overall trends, improving stability and inference especially when data are sparse within some groups.

Complete Pooling



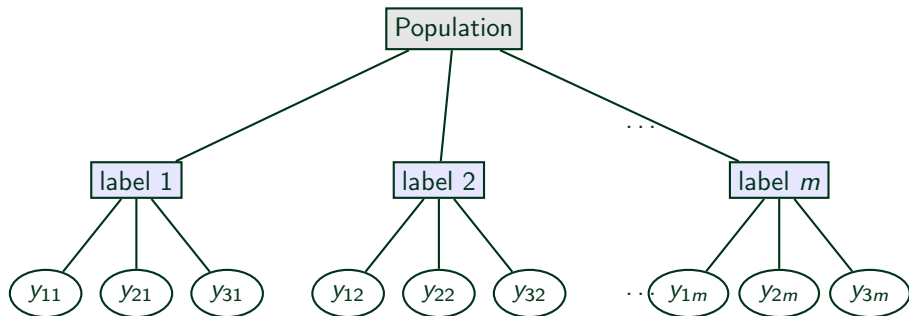
The diagram above represents **complete pooling**, where all observations $y_1, y_2, \dots, y_n$ depend on a single population level parameter, there are no group-specific parameters involved.

No Pooling



In a **no pooling** model, each group has its own parameter θ_i and generates multiple observations y_{ij} independently of other groups. There is no shared structure across groups.

Partial Pooling (Hierarchical)



In **partial pooling**, each observation y_i has its own parameter θ_i , but these are drawn from a common population distribution. This allows for sharing information across groups, leading to more stable and balanced estimates.

Why Hierarchical Models?

Motivation for Hierarchical Data Collection

- It is often more practical and cost-efficient to collect multiple observations within fewer groups than single observations across many.
- Hierarchical data reveal:
 - **Within-group variability:** e.g., consistency in a hotel's yearly performance.
 - **Between-group variability:** e.g., differences in average performance between hotels.

Illustration with labels: Though all models use the same data:

- Complete pooling hides group differences.
- No pooling captures differences but overfits small data.
- Hierarchical (partial pooling) balances individuality and shared strength.

Hierarchical Estimation in Tourism: A Colombian Example

Context: We want to estimate the average daily spending of tourists in three cities: Cartagena, Medellín, and Bogotá. We survey **5 tourists per city**.

Modeling Strategies:

- **Complete Pooling:** Ignores city structure. Combines all data into a single estimate of tourist spending.
- **No Pooling:** Estimates a separate mean for each city using only its own data. Risk of overfitting when sample size is small.
- **Partial Pooling (Hierarchical):** Each city's estimate is adjusted ("shrunk") toward the overall mean. Balances local and global information.

Goal: Show how hierarchical modeling improves estimation by sharing information across structurally related groups (cities), especially when observations are limited.

Tourism Spending Model: Complete vs No vs Partial Pooling

Objective: Predict total tourist spending as a function of length of stay across three cities: Cartagena, Medellín, Bogotá.

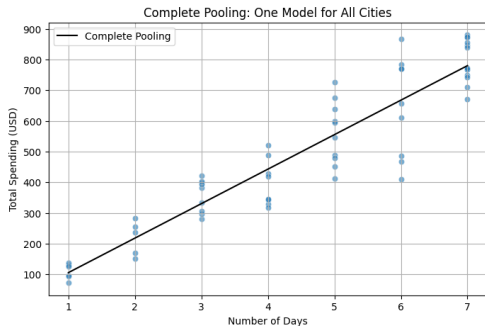
Variables:

- `days` – Number of nights spent
- `total_spending` – Total money spent = `daily` × `days`
- `city` – Categorical grouping variable

Modeling Approaches:

- **Complete Pooling:** One regression line for all cities.
- **No Pooling:** One regression line per city, independently.
- **Partial Pooling:** One model with city-specific effects drawn from a common distribution (hierarchical).

Complete Pooling: One Model for All Cities

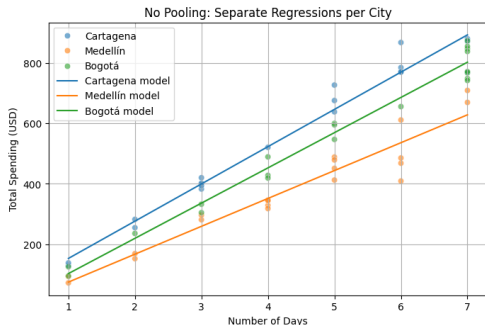


Description:

- A single regression line is fitted across all data.
- Ignores group (city) differences.
- Assumes all tourists behave the same regardless of location.
- Simple and stable, but can underfit group-specific trends.

Interpretation: Useful when group differences are small or data is sparse, but may overlook important heterogeneity in local behaviors.

No Pooling: Separate Models per City

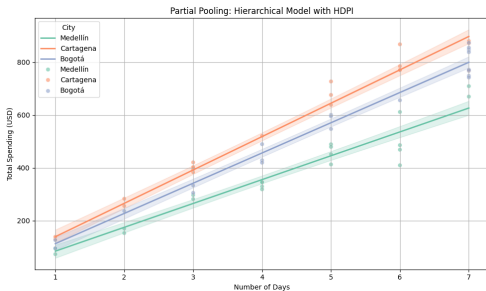


Description:

- Fits an independent regression line for each city.
- No information is shared across groups.
- Captures group-specific behavior accurately.
- Risk of overfitting, especially with limited data per city.

Interpretation: This approach models each city as entirely independent, which works well when group sizes are large and behaviors are truly distinct.

Partial Pooling: Hierarchical Model

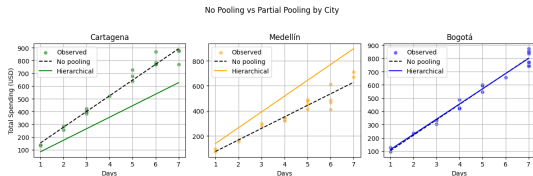


Description:

- Each city has its own regression line, but parameters are informed by a shared distribution.
- Balances flexibility and stability by combining local and global information.
- Allows borrowing strength across groups, reducing overfitting.
- Captures both group-level (city) and population-level patterns.

Interpretation: Partial pooling shrinks estimates toward the global mean, especially useful when group data is limited or noisy.

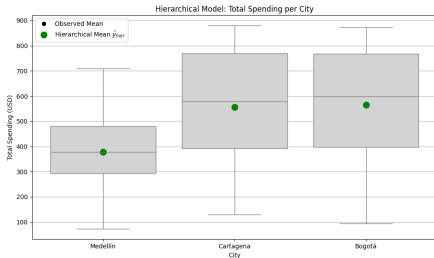
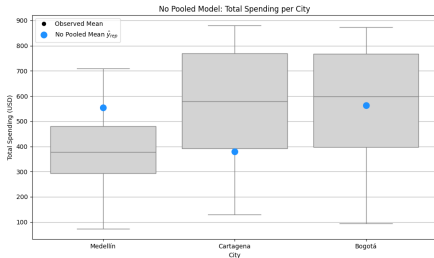
Comparing No Pooling vs Partial Pooling



Why does the hierarchical (partial pooling) line look different in the two visualizations?

- Both plots are based on the same Bayesian hierarchical model.
- The first plot (not shown here) uses the **mean prediction from each posterior draw**, combining thousands of samples, and includes **credible intervals (HDPI)**.
- The plot above uses only the **posterior mean of the parameters** (μ_i , β_i for each city) to create the hierarchical line.
- Therefore, the difference arises from the **use of full posterior draws** vs a **single point estimate** (posterior mean), which smooths the line and reduces uncertainty.
- The full posterior approach captures *uncertainty across draws*, while the plot above only reflects *central tendency*.

Comparing Posterior Predictive Means: No Pooling vs Hierarchical



- **No Pooling Model (left):** Fits a separate model for each city. The predicted means ($\hat{y}_{rep}$) are more affected by local noise and extreme values.
- **Hierarchical Model (right):** Performs *partial pooling* across cities. The predicted means ($\hat{y}_{hier}$) are shrunk toward a global average, reducing the influence of extremes.
- The hierarchical model mitigates overfitting and provides more stable estimates when data is scarce within groups.

Complete Pooling Model

Model Assumption: All cities share the same intercept and slope.

$$y_i = \mu + \beta \cdot \text{days}_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Description:

- Assumes a single linear relationship for all observations, ignoring city-specific differences.
- Most efficient when groups (cities) are homogeneous.
- Can lead to bias if group-level heterogeneity is important.

No Pooling Model

Model Assumption: Each city has its own intercept and slope.

$$y_{ij} = \mu_j + \beta_j \cdot \text{days}_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

Description:

- Fits separate regressions for each group (city).
- Captures group-specific effects well.
- May overfit if data is sparse in some groups.

Partial Pooling: Hierarchical Model

Model Assumption: Group-specific parameters are drawn from a common distribution.

$$y_{ij} = \mu_j + \beta_j \cdot \text{days}_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

$$\mu_j \sim \mathcal{N}(\mu_0, \sigma_\mu^2)$$

$$\beta_j \sim \mathcal{N}(\beta_0, \sigma_\beta^2)$$

Description:

- Allows each city to have its own parameters while sharing strength across cities.
- Balances between flexibility (like no pooling) and generalization (like complete pooling).
- Reduces overfitting and improves estimation in small-sample groups.

Model Comparison: New City with One Observation

Can we predict for a new city with only one data point?

No Pooling (Separate Regressions)

- Each city has its own independent regression.
- Parameters μ_j and β_j are not shared across cities.
- **Problem:** A new city (e.g., Cali) has no estimated parameters.
- **Cannot predict** without retraining the model on Cali.

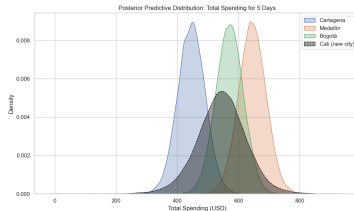
Partial Pooling (Hierarchical Model)

- City-level parameters are drawn from common distributions:

$$\mu_j \sim \mathcal{N}(\mu_0, \sigma_\mu), \quad \beta_j \sim \mathcal{N}(\beta_0, \sigma_\beta)$$

- **Benefit:** Even with just one observation, the model borrows strength from other cities.
- **Can predict** for Cali using the hierarchical prior.

Posterior Prediction for a New City (Partial Pooling)



Posterior Predictive Distributions for Total Spending over 5 Days

- This graph shows the predicted total spending for a 5-day stay in four cities: Cartagena, Medellín, Bogotá, and Cali.
- The first three cities were used in the training data and have many observations.
- Cali, on the other hand, is a new city with only one observation.
- **Partial pooling** uses the hyperpriors to estimate Cali's trend, shrinking the prediction toward the overall population average.
- This approach provides a reasonable estimate for new groups with little or no data, unlike the *no pooling* model, which cannot estimate for unseen categories.